Int. J. Production Economics 173 (2016) 66–79

Contents lists available at ScienceDirect

Int. J. Production Economics

journal homepage: www.elsevier.com/locate/ijpe

A maturity model for enterprise risk management

Fábio Lotti Oliva*

University of São Paulo Business Administration, Avenida Professor Luciano Gualberto, 908, Office C27, 05508-010 Sao Paulo, SP, Brazil

ARTICLE INFO

Article history:
Received 7 June 2015
Accepted 9 December 2015
Available online 19 December 2015

Keywords:
Enterprise risk analysis
Enterprise risk management
Supply chain
Enterprise risk management maturity
Supply chain risk management maturity model

ABSTRACT

The enterprise risk management has been a recurring theme on the agenda of organizations. The competition is increasingly established among the supply chains of organizations. In this sense, it is appropriate that this research aims to analyze the enterprise risk management in the supply chain of Brazilian companies. As a conceptual framework, it has been adopted three theoretical pillars, namely, New Institutional Economics, Supply Chain and Enterprise Risk Management. The research was divided into three stages: interviews with experts on the subject, survey with managers of large Brazilian companies, and validation of the proposals from the analysis of the results with the same experts. The analysis of results was established mainly by the use of multivariate statistical techniques such as correspondence analysis, factor analysis, cluster analysis and multinomial logistic regression. As a result of the study, it was presented a proposal of model for an enterprise risk analysis, as well as a proposal of model for analysis of the level of maturity in enterprise risk management in the supply chain of large Brazilian companies.

1. Introduction

In the corporate world, the attention to events that may compromise the completion of corporate actions largely occupies the agenda of executives. The fierce competition, the bargaining power of clients, the dependence on suppliers, the constant demand for innovation, changes in the regulatory environment, the new expectations of society, make the organizational environment more complex. In contrast, companies seek strategies to raise their internal complexity in order to combat the external complexity that increases gradually. In the context of this research, it is worth noting the increasingly intense adoption of techniques relating to enterprise risk management as a way to ensure the achievement of organizational goals (Mintzberg et al., 2005; Espejo, 1993; Espejo et al., 1996; Damodaran, 2008; Louisot, 2010).

Given the complexity of the organizational environment, it appears that in addition to the dispute between industry agents, clients, suppliers, competitors and new entrants, it is clear that the dispute is established as a competition between supply chains, that is, the aggregation of agents with common goals compete for resources, markets and professionals with other supply chains (Chopra and Sodhi, 2004; Manuj and Mentzer, 2008).

The enterprise risk analysis as an essential part of enterprise risk management has been intensively developed by universities, consulting firms, companies and other organizations. Models and techniques proliferate aiming to incorporate the risk assessment element in the organizational culture. Risk management must permeate the organizational processes (COSO, 2004; ISO 31000, 2009; RIMS, 2006; Ernest and Young, 2012).

According to the executives, studies show that the main risks concentrate on the risks present in the relationships with agents in the business environment. According to the international survey conducted by the AON, a global consulting focused on enterprise risk management, present in more than 120 countries, in 2009, with over 320 executives, showed that the most feared risk by major corporations is the risk of damage to image or to reputation. It is known that reputation is a value built over time with several important agents within the scope of operation of the organization (AON, 2012).

Another important point of consideration lies in the understanding of risk management practices in large organizations. Risk management plays a significant role that only transcends risk analysis (Thun and Hoenig, 2011; Blome and Schoenherr, 2011; Mikes, 2009; Miller and Lessard, 2001). The communication, treatment and monitoring of enterprise risks are some of the tasks inherent to risk management. There has been an effort from the organizations in conceiving a common planning to guide risk management, however, the way how these models are applied and the particularity of each organizational context may reveal unique and distinctive practices that are not described in general manuals. Nevertheless, organizations seek a continuous evolution of their processes, which can also be found in risk management. Thus, knowing the best practices and identifying at what stage the

* Tel.: +55 11 3091 5982.
E-mail address: fabiousp@usp.br

http://dx.doi.org/10.1016/j.ijpe.2015.12.007
0925-5273/

organization is with respect to enterprise risk management have been a desire of companies, consulting firms and other organizations concerned with the risks of their environment of operation (COSO, 2004; ISO 31000, 2009, RIMS, 2006).

Therefore, it is important to incorporate in the theoretical models the analysis of the relationships with agents in the organizational environment. The enterprise risk analysis should consider a scope beyond the boundaries of the organization, that is, the enterprise risk analysis of the organization should include the enterprise risk analysis of its supply chain.

Some authors argue that there are important gaps in supply chain risk management (SCRM) knowledge, for example, clear definition and delimitation about SCRM (Juttner, 2005; Neiger et al., 2009; Zsidisin, 2003; Manuj et al., 2014; Aqlan and Lam, 2015a). Considering supply chain risk management as an important theme emerging from a growing appreciation for supply chain risk by practitioners and by researchers, a distinctive research presents three gaps pertinent to future investigation in SCRM management: no clear consensus on the definition of SCRM; lack of commensurate research on response to supply chain risk incidents; and a shortage of empirical research in the area of SCRM (Sodhi et al., 2012).

Considering supply chain risk management in large company, the critical problems for firms nowadays consist of analysis, evaluation and monitoring of risk in their processes. Poor design and manufacturing decisions, losses in inventory, quality problems in the outsourcing process, cultural management problems are some consequence due to lack of a better supply chain risk management (Manuj et al., 2014; Hult et al., 2010; Lee, 2004; Norrman and Jansson, 2004). In mid of 2012, Deloitte surveyed 600 executives at manufacturing and retail companies, located in North America, Europe and China, to understand their perceptions of their risks. The main results are: (1) 45% understand that their supply chain risk management programs were somewhat effective or not effective, (2) 33% used risk management to provide a proactive and strategic management of their supply chains (Deloitte, 2013).

Based on the problem described above, the purpose of this study is: the analysis of enterprise risk management in the supply chain of large Brazilian companies. Therefore, we intend to identify the main enterprise risks, identify the importance of each enterprise risk in the relationship between company and agent of its supply chain, identify the practices in enterprise risk management. Therefore, as primary contributions this study proposes an enterprise risk analysis model in the supply chain and a maturity level assessment model with respect to enterprise risk management.

The justification for conducting an academic research resides in identifying the importance, contribution, feasibility and originality of the study which reveal its quality (Cooper and Schindler, 2001; Denzin, 1970; Hair et al., 1998). In this research, the importance is established by the need that businesses have to identify and manage its enterprise risks, which go beyond the internal risks and involve other actors of its value chain, considering the most competitive business environment (Manuj et al., 2014; Hult et al., 2010; Lee, 2004; Norrman and Jansson, 2004; Deloitte, 2013). Considering the lack of empirical research on supply chain risk management (Sodhi et al., 2012; Juttner, 2005;.Manuj et al, 2014), the contribution and originality of this research are declared by main objetives which propose an enterprise risk analysis model in the supply chain and a maturity level assessment model with respect to enterprise risk management. The feasibility study should be ensured by adopted methodological procedures. Triangulation, reliability in quantitative research, the validity of questionnaires and investigations are present in this study and are discussed in the methodology (Cooper and Schindler, 2001).

2. Theoretical framework

The theoretical framework adopted in this research was established on three pillars. The conceptual basis must support the development of research tools and the development of the analysis of results. The administrative theory consists of concepts, classifications, relationships and models. Thus, we sought to develop a conceptual framework based on the New International Economics, Enterprise Risk Management and Supply chain. Jointly, it is understood that the three conceptual pillars have contributed to appease the conceptual completeness required for the development of this research.

2.1. New institutional economics

The Institutional Economics is dedicated to the study of institutions, organizations and their interactions in the corporate environment. With a systemic approach, it comprises social, cultural, behavioral and historical aspects to explain the economic relations in society. The classical authors of this section of the economy are: Veblen, Commons and Mitchell. As a point in common, the authors present studies that focus on institutions as elements that order the economic relations between individuals and society. A contrast to the Classical Economics, which addresses the balances of long-term markets driven by rationality and profit maximization of their agents. Considering the Institutional Economics efficient to describe the institutions, but little adherent to explain the economic relations between the agents, the New Institutional Economics emerges with the main texts consisting of the academic exponents: Coase, Williamson and North. In order to increment and adapt to the new demands of reality, instead of rejecting the precepts of the neoclassical theory as did the former institutionalists, the new institutionalists have brought new conceptual elements to the discussion. Coase presents concepts related to transaction costs. Williamson discusses the issue concerning the specificity of assets and the opportunism of agents. North reveals the important concept that institutions order the relations between organizations that seek to obtain marginal gains resulting from variations in relative prices (Nee, 2003; Barzel, 2002; Ménard, 1995, 1997, 2000).

Thus, the conceptual triad proposed by Coase, Williamson and North are revealing for understanding the current economy. Coase (1937) proposes that the costs of the firm's activity transcend the production costs, going beyond the acquisition costs of inputs, processing and sales. The costs for coordinating agents of the economic system to carry out the primary activities of the organization become important in a more complex environment. It is therefore considered that transaction costs consist of costs of information gathering, negotiation and business establishment. In order to qualify these transactions, Williamson (1985) propose two important concepts to the NIE. The opportunism of agents, which reveals the importance of business coordination and that the sizing of its costs can no longer be disregarded, in such a way that they increasingly appear as apparent costs. North (1990) reveals in his own and intense way the importance of institutions as entities that order the relations between agents in the economic environment. Institutions are responsible for the institutional framework comprised by formal and informal rules that govern the behavior of the economic agents, usually grouped in organizations due to common goals.

2.2. Enterprise risks

Enterprises transact with customers, suppliers, government and other organizations. The process of purchasing raw materials and other inputs, production and distribution to markets follows a logical sequence within standards set by society, as well as some non-rational actions of some agents. The environment consists of all relevant forces beyond organizational boundaries organizations are subject to legal changes, changes in social behavior, economic changes, which are often patients in this process, ie, little can influence. For example, companies and their value chain can have their economic activities affected by a tax change, an exchange rate change or a legal challenge filed by the government. On the other hand, companies can react more vigorously forward manner to changes in the relationship with its customers, the competitive design or in the relationship with its suppliers. For example, positive actions or negative actions of employees of a company or other agent in the value chain can affect the perception of the company's image for some time and may even in the most acute cases change the perception of the company's reputation. Many can even act more intensely in their environment and design the future with key stakeholders. In this way, environmental forces affect the objectives of organições thus they are sources of business risks (Bateman, Snell, 2012; Hitt et al., 2014; AON, 2012; Gaultier-Gaillard, Louisot and Rayner, 2009; Kim and Mauborgne, 2004; COSO, 2004).

Considering the risks to which organizations are subject, the literature handles the subject as corporate risks. In this context, enterprise risk is considered to be a measure of uncertainty and includes factors that may facilitate or prevent the achievement of the organizational goals (IFAC, 2001). FERMA (2003), Federation of European Risk Management Association, European association that promotes and disseminates the importance of enterprise risk management in the organizations, defines risk as a combination of the probability of an event and its consequences to the organizations. Being one of the most important international organizations that studies and promotes the importance of enterprise risk management, the mission of COSO, Committee of Sponsoring Organizations of the Treadway Commission, is to ensure the leadership of thought on the subject, developing models and guidelines on risk management, internal control and fraud prevention to improve the organizational performance. For COSO (2004), enterprise risk is represented by the possibility of occurrence of an event and that it will adversely affect the achievement of organizational goals.

Considering an applied interesting research, Deloitte conducted a study from 1994 to 2003, it identified the risks and classified them into four categories: strategic risks, operational risks, financial risks and external risks. And out of these 100 companies, 66% experienced some kind of strategic risk, 61% experienced some kind of operational risk, 37% experienced some kind of financial risk, and 62% experienced some kind of external risk (Deloitte, 2005).

Over the past decade, enterprise risk management has assumed a greater importance. In this period, there was a series of scandals that caused financial losses to clients, suppliers and investors directly connected to companies that operated outside the ethical precepts established, in an indirect manner, the destabilization of the institutional pillars led to major losses to the entire society. The search for a renewed corporate governance that would establish new standards, processes, rules and transparency for all agents, brought in its essence a demand for models and methodologies for enterprise risk management that could become a common benchmark to the organizations and their stakeholders (COSO, 2004, Louisot, 2010; Hoyt and Liebenberg, 2011).

Considering the current practices accepted in business management, a first attempt of ordering risk management occurred in 1994 with the publication of the report Enterprise Risk Management-Integrated Framework by the Committee of Sponsoring Organizations of the Treadway Commission (COSO, 2004). As a result, given the success of this conceptual model, several other entities and other proposals began to emerge to better manage enterprise risks (FERMA, 2003; IFAC, 2001; ISO 31000, 2009; IBGC, 2007). It is worth mentioning the publication of the international standard ISO 31000 as a second wave of proposals to standardize the practices of enterprise risk management. The proposal of ISO 31000 (2009) comprises the presentation of a risk management architecture consisting of principles, structure and process, in such a way that the standard provides a conceptual model and a methodology to be applied in corporations. These principles, structures, processes and models promote a decreasing financial and operating losses, reducing inventory losses, improves supplier relations and reducing external capital dependences. Futhermore, whereas practices of management, it is understood that the good practices of Enterprise Risk Management imply in creating synergies between risk management activities and increasing risk awareness which facilitates better operational, tactical and strategic decision-making (Beasley et al., 2005; McShane et al., 2010; Hoyt and Liebenberg, 2011; Tang, 2006a; Paape and Speklé, 2012).

A global survey conducted by Ernst & Young with over 500 interviews revealed a close relationship between the maturity level in risk management and the financial performance of organizations. A prominent result among those presented by the survey showed that, by considering the ranking of companies with a higher maturity level in risk management, it can be seen that 20% of best rated companies report a financial performance (EBITDA) three times higher the financial performance of worst rated companies in the ranking (Ernest & Young, 2012).

Considering practical and conceptual gaps in Enterprise Risk Management-ERM, some academics and professionals argue that some important gaps which demanding researches to promote more evidences are: (1) the benefits from the implementation of ERM in business; (2) the technical and cultural barriers in the implementation of ERM in business, (3) evaluate if the ERM increase Firm Value (Beasley et al., 2005; McShane et al., 2010; Hoyt and Liebenberg, 2011; Sydow and Frenkel, 2013).

Similarly, assuming that all companies face organizational risks of some kind and that they increasingly demand a formal structure to deal with them, Hillson (1997) proposes a risk maturity model based on four stages: naive, novice, normalized and natural. This model proposes the assessment of the organizations' performance according to the requirements, culture, process, experience and application.

The risk maturity model proposed by RIMS (2006) consists of five stages, ad hoc, initial, repeatable, managed, leadership. The assessment of the level to which the organization belongs consists of the identification of the degree of compliance with seven core attributes established, namely, ERM-based management, ERM process management, risk appetite management, root cause discipline, uncovering risks, performance management, business resiliency and sustainability.

2.3. Risks in the supply chain

Considering the supply chain of a company as the set of all activities of value to the agents involved in its business, it can be seen that competition and cooperation can coexist between agents who share a myriad of converging and diverging interests. The alignment of interests of the agents involved can be considered as the optimal point of business management, but one cannot deny the possibility of reversal in this management. Agents have free will, the interests change, or are not clearly stated, the business environment is dependent on various forces that compose and recompose, and thus opportunism may settle in any relationship and the optimal point may be lost (Porter, Millar, 1985; Porter, 2004, 2008;

Shank and Govindarajan, 1993; Tang, 2006a; Christopher and Holweg, 2011; Habermann et al., 2015).

A proposal with this view was formulated by the IMA (1997), the institute proposes that the relationship between strategy and value creation can be established when management primarily promotes the creation of value for shareholders, and secondly, when management promotes the satisfaction of all other agents involved in the business, provided that it contributes to the creation of value for shareholders. In this sense, enterprise risk management may help maximize the chances of the company to achieve its goals, considering that the enterprise risk is represented by the possibility that an event will occur and adversely affect the achievement of the organizational goals (COSO, 2004).

The academic literature has dedicated effort on the study of the risks involved in the supply chain, extended supply chain, supply chain or production chain, which are often treated as equivalent terms. Given the concreteness and urgency of the relationships suppliers, company and clients, it was found that these related studies to assess the risks involved in the operation and logistics processes to generate value to the client through products and services arouse great interest in the academic and business environment. The risk analysis of the supply chain seeks to identify the main risks that may affect the performance of business processes such as: management of relationships with suppliers, managing of the development of products and services, manufacturing management, management of customer demand, and management of relationships with clients (Ballou, 2004; Chopra and Sodhi, 2004; Christopher and Peck, 2004; Juttner, Peck and Christopher, 2003; Manuj and Mentzer, 2008; Walters and Rainbird, 2004; Walters, 2007; Grey, Shi, 2005; Heckmann et al., 2015).

Based on the principles of the New Institutional Economics, the firm is understood as an instance beyond the set of procedures for the procurement of goods and services for its production processes to generate products and services to be offered to the market. The firm is considered a nexus of contracts, which establishes an order to the internal relationships and external relationships with the agents in the business environment, focusing on efficiency through the reduction of transaction costs and the reduction of production costs (Coase, 1937, 1960; Williamson, 1975, 1991, 1996; North, 1990). Thus, the New Institutional Economics offer a conceptual framework for analyzing the risks involved in the exchange of resources with the organizational environment, that is, the risks in transactions that consist of risks to negotiate, establish and monitor the contracts established between the firm and agents of the organizational environment. On the institutional environment, arise patterns of controlling relationships where trust and power are interrelated such that the risk can best be mitigated (Bachmann, 2001; Aqlan and Lam, 2015b; Olson and Swenseth, 2014; Tang, 2006b; Tang and Tomlin, 2008; Chen et al., 2013; Li et al., 2015).

The risk analysis of the supply chain as an enterprise risk that is only concerned with the risks from the relationships with suppliers, distributors and clients; and likewise, all risks of the company analyzed in the context of the business environment, but without evaluating these same relationships with the agents in the broader context of the business environment, reveals the need to analyze the enterprise risks beyond the boundary of the firm (Manuj et al., 2014, Nooraie and Parast, 2015; Li et al., 2015; Hahn and Kuhn, 2012). Thus, the company's risks can only be understood when analyzed in a systemic manner, that is, the company's risks are the internal risks and the external risks consist of risks posed by other agents to the company and the risks to which the relationships of the company and the agents are subject in the business environment. An emblematic example is the interdependence relationship between the agents of the automobile industry (Thun and Hoenig, 2011). For example, the risks to a company's image or a company's reputation cannot be assessed without considering the risks to the image or the reputations of its suppliers or clients (Gaultier-Gaillard, Louisot and Rayner, 2009).

Therefore, it is understood that the enterprise risk analysis consists of the risk analysis of the environment of value, that is, the analysis of the company's risks in conjunction with its stakeholders in a business environment.

3. Methodology

3.1. Methodological aspects

Considering the research objectives, we elaborated a series of methodological procedures seeking to achieve them. Initially, interviews were conducted with experts on enterprise risk management in large companies. Based on these information and the conceptual model, we developed the research tool to conduct a quantitative research with the companies' managers. Then, we elaborated a data analysis seeking to propose an enterprise risk analysis model and propose a model for assessing the maturity level of the companies with respect to enterprise risk management.

The first stage consisted of a survey with experts who validated the expectations, assumptions and options on the overall purpose of the study, namely, the analysis of enterprise risk management in the supply chain of Brazilian companies. The answers guided the composition of the questionnaire that was applied to managers of large companies.

Then, we developed the survey with managers from large companies. We sought to identify what main risks are assessed on their supply chain and what are the main characteristics in the risk management of these companies. To this end, we developed a questionnaire with closed questions and multiple choice based on the conceptual model and theoretical elements arising from the conceptual framework developed for the subject.

The third stage consisted of the development of a proposal to systematize the process of enterprise risk analysis in the supply chain in which the company participates. In addition, based on data collected in the primary survey, we proposed a model for assessing the maturity level with respect to enterprise risk management.

Aiming to validate the proposals, meetings were held with experts for the presentation of the proposals and collection of criticism and suggestions for improvements, which were incorporated after being evaluated.

3.2. Conceptual model

To meet the stated objectives and answer the research questions, we conducted a literature research on the subjects involved, which supported the development of the conceptual models, research tools and the analysis of results. The literature research focused on the texts addressing the New Institutional Economics, Supply chain and Enterprise Risk Analysis. In this way, developed a conceptual model presented below, which proposes a systemic way to identify enterprise risks in the environment of value of the company.

The business environment consists of various organizations with their relationships that follow a predetermined order. It is in this environment that the company operates. It is divided into macroenvironment and microenvironment. The macroenvironment includes the environmental forces that may influence the company's destination, but that it can barely influence. We highlight the main forces: economic, social, political, technological and environmental. In the microenvironment are the industry forces

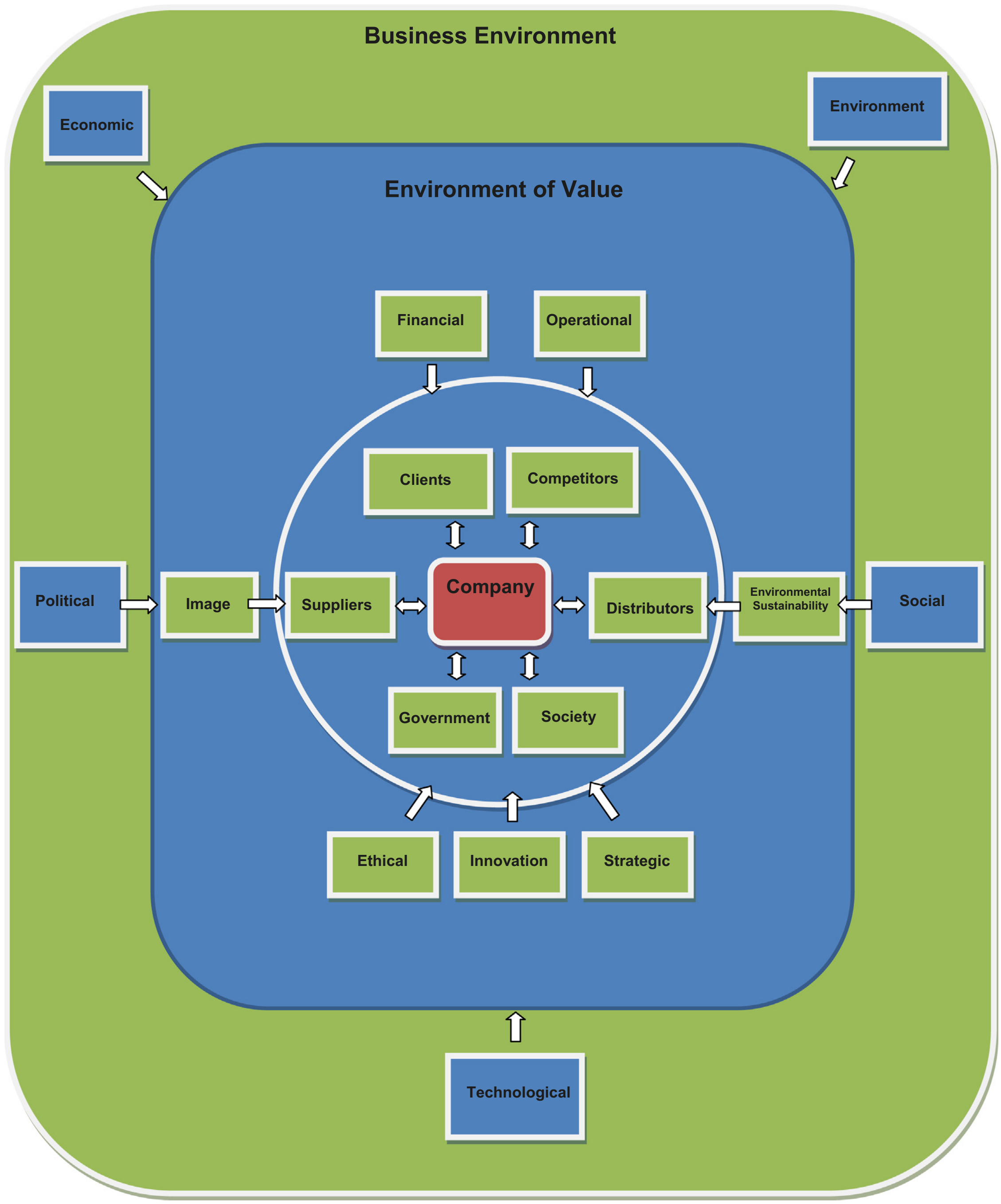


Fig. 1. Enterprise risks in the environment of value.
Source: author (2014).

that can influence the company's destination, which in this case the company can influence more effectively. The main agents are: clients, competitors, suppliers and distributors. The environmental forces in the microenvironment are manifested by the bargaining power, opportunism, innovation and the strategy of its agents (Coase, 1937, 1960, 2014; Williamson, 1975, 1991, 1996; North, 1990; Ménard, 1995; Porter, 2004, 2008) (Fig. 1).

Fig. 1 shows the environment of value, which consists of the main agents that generate value for the company. In this environment, the relationships are established with the intent that both parties are contemplated with what the other part can best offer. Governance levels that maximize the common gain should be sought. The agents of the environment of value are the same agents of the business environment, however, to be considered so, they should maintain or create value for the company. The main agents of the environment of value are: clients, suppliers, distributors, society, government and competitors (Kim and Mauborgne, 2004; Porter, 2008; Shank and Govindarajan, 1993; Christopher and Holweg, 2011; Habermann et al., 2015).

Considering the intention to present a conceptual model for enterprise risk analysis, it is understood that the enterprise risks arise from the internal relationships of the corporation and the company's relationship with the business environment. However, the analysis of the company's risks alone, that is, dissociated from its links of value becomes incomplete and somehow misleading, for example, a company may be in a comfortable condition in relation to the ethical risks and perchance a strategic supplier may be adopting illegal labor practices. Therefore, we advocate that enterprise risks should be analyzed in a systematic manner, considering the company itself and its relationships of value against a business environment with threats and opportunities. Thus, it is understood that the main risks arising from the business environment are originated from the: economic, political, social, technological and environmental events. The risks of the environment of value, where the company is also an agent that influences and is influenced are: financial, operational, image, environmental sustainability, ethical, innovation and strategic (COSO, 2004; ISO 31000, 2009; Ballou, 2004; Chopra and Sodhi, 2004; Christopher and Peck, 2004; Manuj and Mentzer, 2008; Hillson, 1997; RIMS, 2006) (Fig. 1).

The model presented is supported by the theoretical elements of the new institutional economics, supply chain and enterprise risks. The new institutional economics offered a more relational view between the agents, in such a way that it expressed awareness of the importance of assessing the risks to which the corporate relations are exposed, and consequently, how corporate objectives are susceptible to this exposure. The supply chain offered the concept of value creation to the stakeholders. The theory of enterprise risks offered concepts, classifications, relationships and important models to identify the main internal risks of the company, business environment risks and risks from the environment of value.

3.3. Data collection

The primary survey was divided into three field surveys, namely, survey with expert on enterprise risk, survey with enterprise risk managers, and expert validation of the proposed enterprise risk analysis model and the proposed model to assess the maturity level in risk management.

The initial survey with experts consisted of interviews with professionals working in the market, aiming to obtain the key insights on the issue of risk management in major Brazilian companies. The interviews were guided by a script with open questions that sought to explore the following topics: main enterprise risks, main methods to analyze enterprise risks, and best practices in enterprise risk management. We interviewed three managers of large corporations, a manager in the automotive industry, a manager in the food industry and a manager in the electronic equipment industry, who presented their experiences in enterprise risk management in large companies. In addition, we interviewed two consultants in enterprise risk management, who presented their experiences as consultants in several large companies.

With this information, we elaborated the pre-understanding on which concepts of the theoretical framework defined previously could better represent the risk management process of large Brazilian companies. In other words, we defined a preliminary proposal for the systematization of risk analysis and the assessment of the maturity level of risk management, which guided the development of the quantitative research instrument (see Fig. 1).

As for the survey with managers, in the second stage of the research, we carried out extensive research with managers from different areas of the selected organizations. Considering the population with the 1,100 largest Brazilian companies, according to the ranking published by EXAME magazine (2011), consisting of 1000 companies from the non-financial sector plus 100 companies from the financial sector. The non-financial sector companies were ranked by the sales revenue, an indicator of the company's contribution in terms of products and services offered to society in 2010. In terms of sales volume, the 1000 companies earned more than US$ 1.5 trillion. The financial sector companies consist of the 50 largest banks, ranked by net equity, and the 50 largest insurance companies, ranked by the net premiums written.

We defined a random representative sample of companies of size (n), based on the population size (N), sampling error (e), confidence interval (z), and estimated standard deviation, as follows:

$$n=\frac{s^2\cdot N\cdot Z^2}{d^2\cdot(N-1)+s^2\cdot Z^2}=\frac{3{,}34^2\cdot 1.100\cdot 1.96^2}{0{,}50^2\cdot(1.100-1)+3.34^2\cdot 1.96^2}=149$$

s = estimated standard deviation = 3.34;

N = population size = 1.100;

Z = abscissa of the standard associated with confidence level of 95% = 1.96; d = sampling error = 0.50

The original sample consisted of 243 large companies. We managed to conduct the survey in 181 companies, that is, over 25% of the companies did not agree to participate or we were unable to contact a manager capable of doing so. We applied the questionnaire, to an employee of the company with academic and professional experience appropriate to answer questions about enterprise risk management. It is understood that the enterprise risk management must permeate the core processes of the organization, therefore, there was no attempt to limit the survey to professionals who are experts on the subject. Among the questionnaires answered, 168 can be considered valid, that is, 13 questionnaires were discarded due to the lack of information or error of completion.

Considering the survey instrument, in the analysis of enterprise risks were presented the main risks to which the companies surveyed are submitted in relation to the business environment of the agents, based on the conceptual model (Fig. 1). Considering the data collection, which requires the respondent to award a grade 0-10 to declare the importance of risk in relation to the agent of its business environment, joined by multivariate correspondence analysis which most important risks for certain agents. The research instrument presents risks divided into risks environmental risks and risks of the value chain. Note that the assessed risks were declared for each agent involved in the company's value system. Regarding the assessment of the maturity level in enterprise risk management of the companies surveyed, developed 18 questions about corporate risk management, considering the variables on the maturity stages in risk management based on the literature (Hillson, 1997; RIMS 2006; COSO, 2004; ISO 31000, 2009; IBGC, 2007). As a result, we developed a multivariate analysis thread for this purpose, namely, factor analysis, cluster analysis and multinomial logistic regression.

The third field research consisted of the validation of the results achieved in the analysis of data with experts. We sought to conduct meetings with the same managers and consultants who collaborated with the pre-understanding of the research problem, where we presented the enterprise risk analysis and maturity level assessment models in enterprise risk management. Furthermore, the results were presented to two professors who research and teach on issues related to enterprise risk management. During the meetings it was possible to collect some improvements that have been incorporated. In general, the contributions were not structural, in such way that it can be considered that the proposals are consistent with the understanding of the respondents.

4. Analysis of results

4.1. Enterprise risk analysis

Enterprise risks may be of different nature and may have different degrees of importance to each agent. A part of the field research with managers of large companies was intended to reveal the importance of each risk for each relationship with the agents of their environment of value. Based on the knowledge of each manager, we asked them to assign a score from 0 to 10 for each enterprise risk and for each agent arranged in a matrix, therefore, the manager should assign, according to their understanding, for example, what is the importance of political risk for their relationship with the government.

Considering the scores assigned by the managers, we developed a correspondence analysis two dimensions, namely, events and agents. The events were divided into two categories, business environment and environmental of value. In the business environment, we presented the events related to the macroenvironment: economic, environmental, political, social and technological. In the environment of value we presented the events related to the following aspects: financial, operational, image, environmental sustainability, ethical, innovation and strategic. The agents are organizations that have an important relationship with the company, they are organizations deemed important in the maintenance or creation of value to the company, that is, the organizations present in the environment of value the company. The agents considered in this research were: the company, clients, suppliers, competitors, distributors, government and society.

With regard to the validity of using the correspondence analysis technique, we rejected the hypothesis that the categorical variables, events and agents, are independent at a level of significance of 5%. In addition, we calculated the value of β to confirm the dependence of the variables (Hair et al., 1998; Pestana and Gageiro, 2000).

$$\beta = \frac{\chi^2 - (l-1)(c-1)}{\sqrt{(l-1)(c-1)}} = \frac{435.69 - (12-1)(7-1)}{\sqrt{(12-1)(7-1)}} = 45.54$$

X^2 : value of chi − square

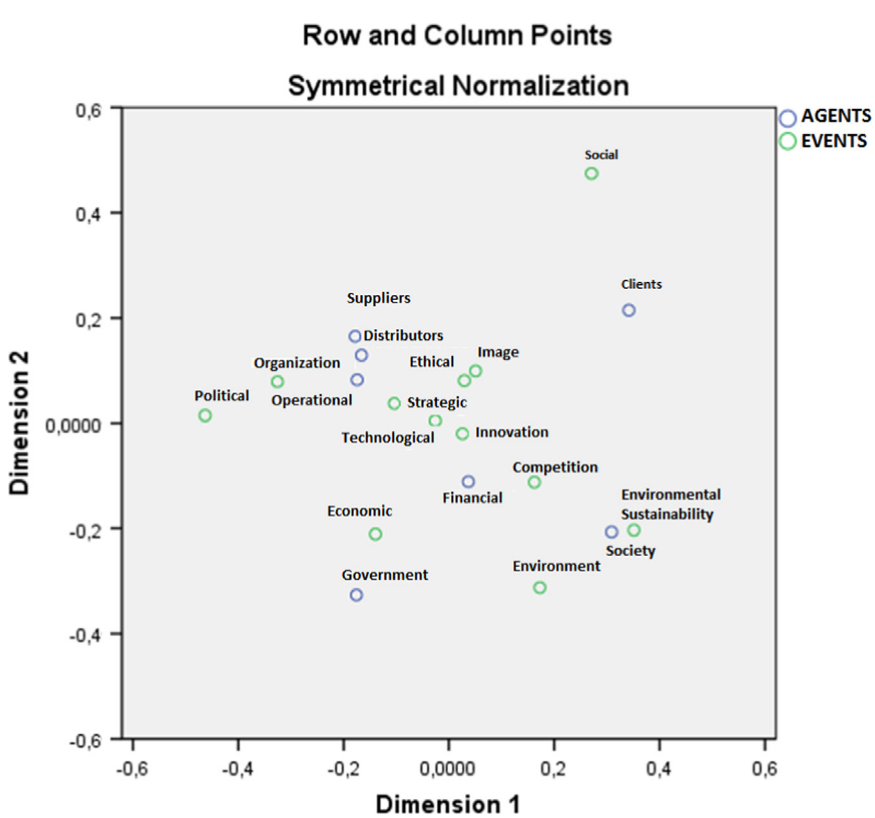
l : number of rows of the contingency table

c : number of columns of the contingency table

In other words, where $\beta > 3$, it is considered that the categorical variables, events and agents, are dependent at a risk of 5%.

The multivariate technique provides a perceptual map in two dimensions where it is possible to identify the closest events to the agents, that is, to identify which enterprise risks are considered most important to each agent of its environment of value. Graph 1

Considering the graphic dispersion of events and agents, initially, it can be seen that the enterprise risks arising from environmental sustainability and environment issues are more sensitive to society, which seems to be consistent with the latest desires of society; there are various efforts from non-governmental organizations engaged in changing the human attitude with regard to the respect for nature (Hofmann et al., 2014; Veiga, 2005). For competitors, it appears that the most important risks in the relationship between the company and competition are the enterprise risks: financial, innovation and technological. The difference in prices charged by competitors appeared as an important source of risks. Opportunistic relationships are established when the ordering of the institutional environment is not fully established or is at least acceptable by the main agents (North, 1990; Ménard, 1995, 1997, 2000). Both the innovation from competition and the technological changes in the business environment are considered relevant enterprise risks to the company, which must be constantly alert to these types of events that may dramatically change the relationship of power between the agents of its environment of value (Kim and Mauborgne, 2004). Regarding the most important risks to the company, the operational risks and strategic risks stand out. Corporate risks arising from the lack of coordination of the supply chain can lead to serious problems in the acquisition of raw material, production and delivery of products and services to customers (Li et al., 2015; Habermann et al., 2015). Strategic risks, for example, arising from the managers' lack of ability to understand the changes in the business environment, may cause the corruption of the company's value or the complete failure of its operation. For



Graph 1. Perceptual map of the correspondence analysis.
Source: Author (2014).

instance, bets on certain characteristics of new products that require large investments and which have the chance of being rejected by society may lead the company to an adverse financial situation or damaged image possibly without recovery (Chopra and Sodhi, 2004; Louisot, 2010; Nooraie and Parast, 2015).

Given the data collection method, which provides the relativity of the importance assigned by the managers to the company's risks and its relationships with key agents of its environment of value, Table 1 below shows the risks in order of importance given the proximity observed in the perceptual map.

4.2. *Evaluation of the level of maturity in enterprise risk management*

The questionnaire was administered to a sample of 243 managers from the population of the 1,100 largest Brazilian companies according to the ranking of EXAME (2011). With the database organized, we conducted a factor analysis, data reduction, aiming to identify the important factors for the explanation of the phenomenon studied. Then, we conducted a cluster analysis aiming to identify the possible groups with distinct approaches to enterprise risk management. Based on the categorization achieved, it was possible to develop a multiple logistic analysis, it allows the identification of which variables are the most important to assess whether the company belongs to that cluster. The predictive power of this analysis can be noted by the use of the equations associated with each cluster. In such a way that we come up with a way to assign the level of maturity of the company with respect to the enterprise risk management (Fig. 2).

4.2.1. *Analysis of the explanatory factors of enterprise risk management*

Initially, we performed the analysis of the correlation matrix of the variables. According to Hair et al. (1998), there should be a large number of correlations higher than 0.30. In this case, it is worth noting that out of the 324 positions of the matrix$_{18\times18}$, only 34 positions indicate value lower than or equal to 0.30, that is, 10.49% of the positions. However, it is worth mentioning that these positions arise only from variable 2, which have low correlation with other variables. Since this variable was eliminated from the analysis, there is no position lower than or equal to 0.3, only 12 positions are lower than or equal to 0.4, that is, 3.70% of positions is lower than or equal to 0.40. Therefore, considering this aspect, the factor analysis appears to be appropriate for use. In addition, we calculated the Kaiser-Meyer-Olkin (KMO) statistics, which indicates the degree of partial correlation between the variables. In this sample, it was found that the KMO value is equal to 0.955. According to Pestana and Gageiro (2000), for samples with KMO value lower than 0.60, the factor analysis may not be adequate. Thus, for the sample of the research, the use of factor analysis is appropriate. Furthermore, we performed Bartlett's sphericity test, which consists in testing the hypothesis that the correlation matrix of the variables is the identity matrix. By considering that the *p*-value is equal to 0.000, we eliminate the hypothesis that the correlation matrix of the variables is the identity matrix, therefore, the use of the factor analysis for this sample is appropriate. Table 2 below shows the final values of the validity statistics of the use of the factor technique.

The factor analysis showed that the risk management practices declared by the managers can be classified into four explanatory factors, namely, organization, technicality, transparency and involvement (see Fig. 3). The factor analysis provides groupings of variables, called factors, which have the same numerical behavior with respect to the responses of the survey, in such a way that these variables can be replaced, for the purposes of analysis, by the factor, thus facilitating the understanding of the phenomenon. Obviously, reducing information from the model always cause the reduction of its explanatory power. Considering an appropriate adoption of the factor analysis technique, it is possible to obtain factors with strong explanatory power that simplify the analysis due to the reduction of the elements to be evaluated. Therefore, based on the questions of the interview, the variables associated with the numerical factors proposed by the factor analysis, we assign a meaning to each one, where the first factor was called Organization, which represents how much the company dedicates efforts to produce a structured risk management. The second factor is called Technicality, which seeks to portray how often the company makes use of qualitative or quantitative techniques to support the risk management process in the corporation. The third factor, Transparency, reveals how often the company addresses the subject openly with its collaborators, seeking to involve them in participatory management of enterprise risks. Finally, the last factor is called Involvement, as it is expected to show how much the company is able to engage other agents of its environment of value to make its risk management more efficient and effective.

Table 1
Most Relevant Risks in the Environment of Value.
Source: Author (2014).

Agents	Most relevant risks
Organization	Strategic, Operational, Ethical and Image
Clients	Social and Image
Suppliers	Strategic, Operational, Ethical and Image
Competitors	Financial Innovation, Technological, Ethical and Image
Distributors	Strategic, Operational, Ethical and Image
Government	Economic
Society	Environmental Sustainability and Environment

Table 2
Factor analysis statistics.
Source: Author (2014).

Statistics	Values
KMO	0.955
Bartlett's sphericity test (*p*-value) (level of significance <0.05)	0.000
Minimum, maximum and average commonality	0.729; 0.919; 0.807
Minimum, maximum and average MSA	0.925; 0.976; 0.954
Minimum, maximum and average factor loading	0.554; 0.900; 0.729
Eigenvalues	10.49; 0.98; 0.81; 0.64
Number of selected variables	8
Number of aggregating factors (eigenvalues over=0.5)	4
Variance explained by the aggregating factors	22.46%; 32.37%; 15.07%; 10.84%
Total variance explained by the aggregating factors	80.75%

Fig. 2. Enterprise risk management maturity level data analysis.
Source: Author (2014).

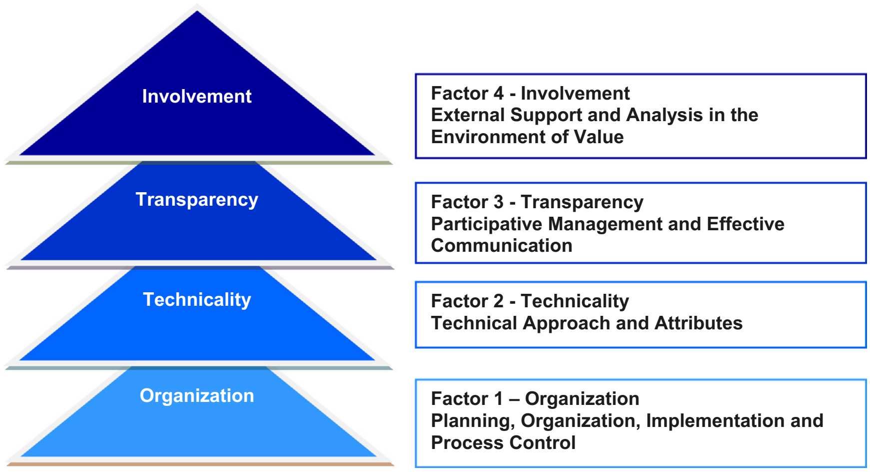


Fig. 3. Explanatory factors of enterprise risk management.
Source: author (2014).

Considering the factors presented, based on a numerical analysis of the data, it is observed that companies have different degrees of organization, technicality, transparency and involvement. A company can be more organized and less transparent with respect to risk management, or may involve more agents of its environment of value in pursuit of a more systemic risk management and possess a low degree of technicality in its risk management. However, in general, it can be seen that there is a certain hierarchy regarding the factors involved in risk management. All companies have some degree of organization, technicality, transparency and involvement. However, for example, the greater the degree of technicality, the greater the degree of organization. The greater the degree of transparency, the greater the degree of technicality and organization. The greater the degree of involvement, the greater the degree of transparency, technicality and organization. That is, apparently, the companies have certain hierarchy with respect to the explanatory factors that classify the enterprise risk management. In the following analyzes this issue will be properly detailed.

4.2.2. Analysis of characteristic groups in enterprise risk management

In this study, we decided to conduct the hierarchical cluster method to identify the ideal number of groups and the initials centroids, and then, based on the identified data, we conducted the non-hierarchical cluster method K-means. This composition of the use of methods of hierarchical and non-hierarchical clusters allows us to obtain the best of each one. The hierarchical method is advantageous when the ideal number of clusters is unknown, which occurs in the vast majority of studies, on the other hand, the method proves to be disadvantageous because its genesis consists of the hierarchical composition of clusters, that is, initially, each element is a potential cluster and the new clusters are established from the previous ones, so that once the elements are grouped, they will remain in the same cluster until the end of the processing. However, the non-hierarchical method necessarily needs a suggested number of clusters and preferably the initial centroids, otherwise, the algorithm will define which information it will use as a starting point. On the other hand, the method proves to be advantageous because the method seeks the choice of the best cluster configuration for the database presented (Hair et al., 1998).

Thus, we started the processing of the hierarchical cluster analysis, establishing as a measure of similarity the Euclidean squared distance measure and as the clustering method the between-groups linkage. This method seeks to create groups with the smallest distance, which is calculated as the average of the individual distances of each element of a group with the elements of another group. Considering the numerical analysis of the Agglomeration Schedule and the visual analysis of the Vertical Icicle and Dendrogram, it can be seen that the ideal number of clusters is 6 groups. The Agglomeration Schedule shows the composition of the groups step by step, so that it is possible to evaluate the distance of the two groups associated with each iteration. The criterion for defining the optimal number of clusters is the verification of the first major difference between the distances of each interaction. In this case, the first major difference occurs between interaction 159 and 160, in the value of 1.481, which consist of the identification of 9 clusters. Considering this large number of groups, it can be seen that the next major variation occurs between interactions 162 and 163, in the value of 1,593, thus providing the optimal identification of six clusters.

Table 3
Number of companies by cluster.
Source: Author (2014).

Cluster	Number of Companies
C1	34
C2	57
C3	16
C4	22
C5	28
C6	11
Total	168

As previously mentioned, following the processing of the hierarchical cluster analysis, we used the non-hierarchical cluster analysis K-means. The K-means method adopts as measure of similarity the Euclidean distance and as the clustering method the nearest centroid sorting, which seeks to minimize the internal variance of the elements of each cluster and maximize the variance between clusters (Pestana and Gageiro, 2000).

After determining the composition of 6 clusters, the processing of the non-hierarchical cluster analysis K-means presented the following configuration of segmentation of the companies surveyed (Table 3).

The analysis of variance table, ANOVA, shows which variables contributed most to discriminate the clusters. Initially, it appears that the four variables, factors obtained in the factor analysis, were significant for the formation of the 6 clusters at a significance level of 5%. Consecutively, it is found that the F statistics, present in the analysis of variance, indicates how much the variable has contributed to the composition of the clusters. The higher the value of the F statistics, the more the variable discriminates the groups. In this case, by order of greatest contribution in the discrimination of the clusters, we have the following variables: Factor 4 – Involvement, Factor 1 – Organization, Factor 2 – Technicality and Factor 3 – Transparency. That is, Factor 4 – Involvement was the variable that contributed the most to discriminate the clusters, $F=42.238$, in such a way that the external variability is greater, 18.901, and the internal variability was the lowest, 0.447. Table 4

By refining the cluster analysis obtained, we sought to calculate the average values of the original variables of the factors for the elements of each cluster, for example, companies in cluster C1 have average values of the original variables of factor 1-organization equal to 4.5294, or companies in cluster C2 have average values of the original variables of factor 4 – involvement equal to 7.4298. It is worth noting that the original variables vary from 0 to 10. There is a relationship between the factors and groups obtained. Cluster C1 shows low values by factors, all lower than 5, in such a way that it was considered a group with insufficient characteristics for a classification of the enterprise risk management. On the other hand, cluster C4 indicates value 5.8182 for factor 1-organization and lower values the subsequent factors. Cluster C5 keeps the average value for factor 1 above 5 and indicates value 6.4063 for factor 2-technicality. Cluster C3 keeps the average values for factors 1 and 2 above 5, and indicates value 5.7917 for Factor 3-transparency. With the same behavior, cluster C2 keeps the average values for factors 1, 2 and 3 above 5, and indicates value 7.4298 for factor 4-involvement. Overall, there is a hierarchy between the clusters, the average values show that the hierarchically superior group maintains higher values for the previous factors and indicates a value explicitly higher to the next factor, for example, cluster C3 has higher average figures for factors 1 and 2 compared to the average values of clusters C4 and C5, in addition, it indicates a greater average value for factor 3, when compared to the average values of clusters C4 and C5 (see Table 5 and Chart 1).

It should be mentioned that cluster C6, which gathers 11 elements, was eliminated because it indicates atypical average values for the consolidation of the behavior intended to be shown. The values of factors 1, 2, 3 and 4 are 4.3030; 3.3523; 4.0303; 6.7727, respectively, values lower than 0.5, except for factor 4, moreover, there is no increasing behavior of the values compared to the factors as presented by the other clusters.

The hierarchical behavior will be more explored with the presentation of the results obtained with the application of the technique of multivariate multinomial logistic regression analysis for the categories represented by the clusters.

Table 4
Evaluation of clusters.
Source: Author (2014).

Variable	Cluster mean square	Error mean square	*F*	Sig.
Factor 1 – Organization	18.448	0.461	39.974	.000
Factor 2 – Technicality	17.290	0.497	34.772	.000
Factor 3 – Transparency	11.346	0.681	16.668	.000
Factor 4 – Involvement	18.901	0.447	42.238	.000

4.2.3. Analysis of the level of maturity in enterprise risk management

The multiple logistic analysis is a multivariate technique that associates a categorical variable Y of p instances with a set of q independent variables X. Among its main statistical attractiveness, we highlight the greater flexibility regarding the prerequisites of its application. The normality and homogeneity of variance of residuals are not required. The technique and the multiple regression allow us to identify what are the key independent variables associated with the categories of the dependent variable Y. In addition, as a distinctive offer, the multiple logistic technique allows the estimation of the probability of occurrence of a specific event (Hosmer and Lemshow, 2000).

$$Y_p = X_1 + X_2 + X_3 + \ldots + X_q$$

Y is a categorical independent variable that can assume p distinct values
X_i are metric or non-metric dependent explanatory variables

The formulas for calculating the probabilities of a company to belong to a certain level of maturity in knowledge

Table 5
Factors and clusters.
Source: Author (2014).

Factor	Clusters				
	C1 – Insufficient	C4 – Contingency	C5 – Structured	C3 – Participative	C2 – Systemic
F1 – Organization	4.5294	**5.8182**	5.3214	8.1875	7.8187
F2 – Technicality	3.5588	3.3807	**6.4063**	7.6563	7.4583
F3 – Transparency	4.0882	2.1515	4.5238	**5.7917**	6.9181
F4 – Involvement	2.9853	3.9773	5.9643	4.1875	**7.4298**
Average	**3.7728**	**3.7118**	**5.7465**	**6.8693**	**7.4181**

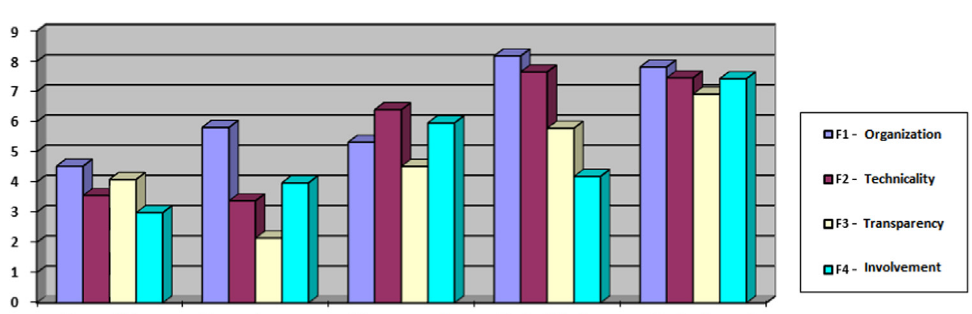


Chart 1. Factors and clusters.
Source: Author (2014).

management are:

$$g(1) = ROR_{y=1,y=1} = \frac{odds(y=1)}{odds(y=1)} = 1$$

$$g(2) = ROR_{y=2,y=1} = \frac{odds(y=2)}{odds(y=1)} = e^{\alpha_2 + \sum_{i=1}^{8} \beta_{2i} X_{2i}} = e^{Z_2}$$

$$g(3) = ROR_{y=3,y=0} = \frac{odds(y=3)}{odds(y=1)} = e^{\alpha_3 + \sum_{i=1}^{8} \beta_{3i} X_{3i}} = e^{Z_3}$$

$$g(4) = ROR_{y=4,y=0} = \frac{odds(y=4)}{odds(y=1)} = e^{\alpha_4 + \sum_{i=1}^{8} \beta_{4i} X_{4i}} = e^{Z_4}$$

$$g(5) = ROR_{y=5,\, y=0} = \frac{odds\,(y=5)}{odds\,(y=1)} = e^{\alpha_5 + \sum_{i=1}^{8} \beta_{5i} X_{5i}} = e^{Z_5}$$

$$P(y=1) = \frac{g(1)}{g(1)+g(2)+g(3)+g(4)+g(5)} = \frac{1}{1+g(2)+g(3)+g(4)+g(5)}$$

$$P(y=2) = \frac{g(2)}{g(1)+g(2)+g(3)+g(4)+g(5)} = \frac{g(2)}{1+g(2)+g(3)+g(4)+g(5)}$$

$$P(y=3) = \frac{g(3)}{g(1)+g(2)+g(3)+g(4)+g(5)} = \frac{g(3)}{1+g(2)+g(3)+g(4)+g(5)}$$

$$P(y=4) = \frac{g(4)}{g(1)+g(2)+g(3)+g(4)+g(5)} = \frac{g(4)}{1+g(2)+g(3)+g(4)+g(5)}$$

$$P(y=5) = \frac{g(5)}{g(1)+g(2)+g(3)+g(4)+g(5)} = \frac{g(5)}{1+g(2)+g(3)+g(4)+g(5)}$$

Considering category 1 as reference, that is, the calculations are based on level 1 of maturity, we can apply the formulas to obtain the probability of a company X to be at a certain level of maturity (see Table 6).

Considering the variables of the research, the components of X assume the respective values of the original variables of the research, therefore, X=(Q1, Q3, Q4, Q7, Q10, Q12, Q13, Q17). Table 7

Note that the original variables Q1, Q3 and Q4 are part of factor 1-Organization, Q7 and Q10 of factor 2-Technicality, Q12 and Q13 of factor 3-Transparency, Q17 of factor 4 – Involvement, are present in the equations of the multinomial logistic model, that is, the four factors deemed important to define the clusters are also present in the equations of the logistic model, through the original variables that comprise them, thus confirming that all four factors are discriminant of the companies surveyed. Therefore, considering the predictive power of the model, the answers to these questions will be important to determine in which cluster a company can be classified, that is, what is its level of maturity in risk management. In addition, the model provides the probability of this company to belong to each of the possible levels of maturity. Thus, considering the 18 original variables, it can be seen that the 8 variables Q1, Q3, Q4, Q7, Q10, Q12, Q13 and Q17 are the most important variables for defining the level of maturity in the risk management of a company.

As for the adhesion of the model, it is found that the significance test of the coefficients of the final model indicates value 0.000, lower than 5%, that is, the model was significant at the level of 5%.

With regard to the variability captured by the model, that is, the explanatory power of the model, it is considered high because all the pseudo R^2 are higher than 94%. The Cox and Snell R^2 is equal to 94.46%, Nagelkerke R^2 is equal to 99.26% and McFadden R^2 is equal to 95.49%.

Further evidence regarding the adhesion of the model is represented by the Likelihood Ratio Test of the model, and it can be seen that out of the 18 variables collected, 8 are significant for the composition of the model equations. The coefficients of the equation are significant at the level of 5%. The test results are lower than 0.000, except for the coefficient of variable Q4 whose value is 0.039.

In addition, with respect to the adhesion of the model, the Goodness-of-fit hypothesis tests appear to be favorable. The null hypothesis Ho that there are no significant differences between the values observed and predicted by the model is not rejected. The p-values of Pearson and Deviance hypothesis test are greater than 0.05. Thus, the multinomial logistic model presented appears to be duly in line with the survey data.

On the other hand, the analysis of the estimated parameters does not indicate that the coefficients of the equations are significant at the level of 0.05. Likewise, the Wald test, where the null hypothesis Ho indicates that the coefficients are equal to zero is not rejected at the level of 0.05. However, some studies show that the Wald test has become unstable in certain situations, suggesting that the adhesion of the model should not only be analyzed by this statistic, but by a set of statistic of adhesion pertaining to the multinomial logistic regression (Hauck and Donner, 1977).

As far as the measures already presented, the percentage of accuracy of the model is another important evidence to evaluate the adhesion of the model presented. It was found that, in general, for all the five categories, the level of accuracy is greater than 92%, which sets a high predictive power of the model. The average percentage of accuracy is 97.5%, that is, out of the 157 companies surveyed, the model predicted the clusters correctly for 153 companies. Table 8

Thus, given that most adjustment statistics reveals that the model has high level of adhesion and the level of overall accuracy

Table 7
Original variables comprising the multinomial logistic model.
Source: Author (2014).

Q1	What is the LEVEL of CONTROL of the organization's risk?
Q3	Are the processes relating to risk Planned, Organized, Executed and Managed rationally?
Q4	Are the processes relating to risk dealt with adequate FREQUENCY?
Q7	Are the risks MEASURED QUANTITATIVELY?
Q10	Is there a CULTURE OF RISK ASSESSMENT in the management of the organization?
Q12	Are there OPEN EVENTS, with ideal frequency, to discuss the management risks of the organization?
Q13	Is the risk management in the organization DECENTRALIZED?
Q17	What is the LEVEL OF EXTERNAL SUPPORT (consultancy, university, partners or other entities) to the enterprise risk management?

Table 6
Equations by clusters.
Source: Author (2014).

Cluster	Equation
C1 (y=1)	$Z_1 = 0$
C2 (y=2)	$Z_2 = -79,82 + 5,44\,X_1 + 3,23\,X_2 + 0,64\,X_3 + 3,29X_4 + 1,23X_5 + 1,17X_6 - 3,78X_7 + 3,67X_8$
C3 (y=3)	$Z_3 = -316,52 + 7,13\,X_1 + 29,02\,X_2 + 8,24\,X_3 + 21,96X_4 + 0,97X_5 - 8,46X_6 - 4,51X_7 - 29,82X_8$
C4 (y=4)	$Z_4 = -46,82 + 15,57\,X_1 + 6,62X_2 - 0,91X_3 + 0,83X_4 - 11,42X_5 - 3,03X_6 - 10,76X_7 + 4,29X_8$
C5 (y=5)	$Z_5 = -29,33 + 0,71\,X_1 - 0,12X_2 - 1,42X_3 + 5,28X_4 + 4,27X_5 - 0,49X_6 - 5,22X_7 + 3,32X_8$

is 97.5%, it is understood that the multinomial logistic model presented is in line with the data collected and given the representativeness of the sample, the model can be extrapolated to the population of companies surveyed.

Considering the distinctive characteristic of the logistic models in providing a calculation of probability for the occurrence of an event, duly represented by the categories of the dependent variable, below we present one example of calculation of such probabilities Table 9.

Table 8
Evaluation of the level of accuracy of the prediction model.
Source: Author (2014).

	Clusters Predicted					
Clusters observed	1	2	3	4	5	Percentage of accuracy
1	34	0	0	0	0	100%
2	0	55	0	0	2	96.5%
3	0	0	16	0	0	100%
4	0	0	0	22	0	100%
5	0	2	0	0	26	92.9%
Average						97.5%

To illustrate the calculations of probability, we assumed variable X in the following instance X_{32}=(7,8,8,6,2,4,5, and 3), which sets the following values:

Table 9
Example 1 of application of the multinomial logistic model.
Source: Author (2014).

Cluster	Z_j	$P(y=j)$
C1 (y=1)	$Z_1 = 0$	$P(y=1)$=00.00%
C2 (y=2)	$Z_2 = 8.17$	$P(y=2)$=00.00%
C3 (y=3)	$Z_3 = 19.26$	$P(y=3)$=00.00%
C4 (y=4)	$Z_4 = 36.95$	$P(y=4)$=99.99%
C5 (y=5)	$Z_5 = -14.60$	$P(y=5)$=00.00%

It can be seen that the company 32 belongs to cluster 4, level 1 of maturity, that is, it is a company, according to its manager, who has a good level of organization, however, it has a low level of technicality, transparency and involvement in enterprise risk management.

Therefore, based on the above, we propose a model for assessing the level of maturity of enterprise risk management with 5 categories, which are revealed by the enterprise risk management characteristics assumed by the companies. In Fig. 4, the categories are presented and right after each one is described, that is, it shows the characteristics that an organization must take to be considered in this category of maturity in enterprise risk management.

Level 1, called *insufficient* enterprise risk management, includes companies that have little awareness of the enterprise risks. There is no physical or conceptual structure dedicated to enterprise risks. The adoption of risk management practices occurs on a non-structured manner.
Level 2, called *contingency* enterprise risk management, involves companies that are aware of the risks to which they are subject. Risk management techniques, tools and methods are roughly used. Risk management is centralized and is characterized by the low involvement of employees in general.
Level 3, called *structured* enterprise risk management, involves companies with a higher degree of organization of processes related to enterprise risk management. There is a more intense use of risk management techniques, tools and methods.
Level 4, called *participative* enterprise risk management, involves companies with high level of awareness and organization with regard to the processes related to enterprise risk management. The risk management is more decentralized. Communication is an integral and important part in risk management. The enterprise risk management is guided by the participation of most employees.

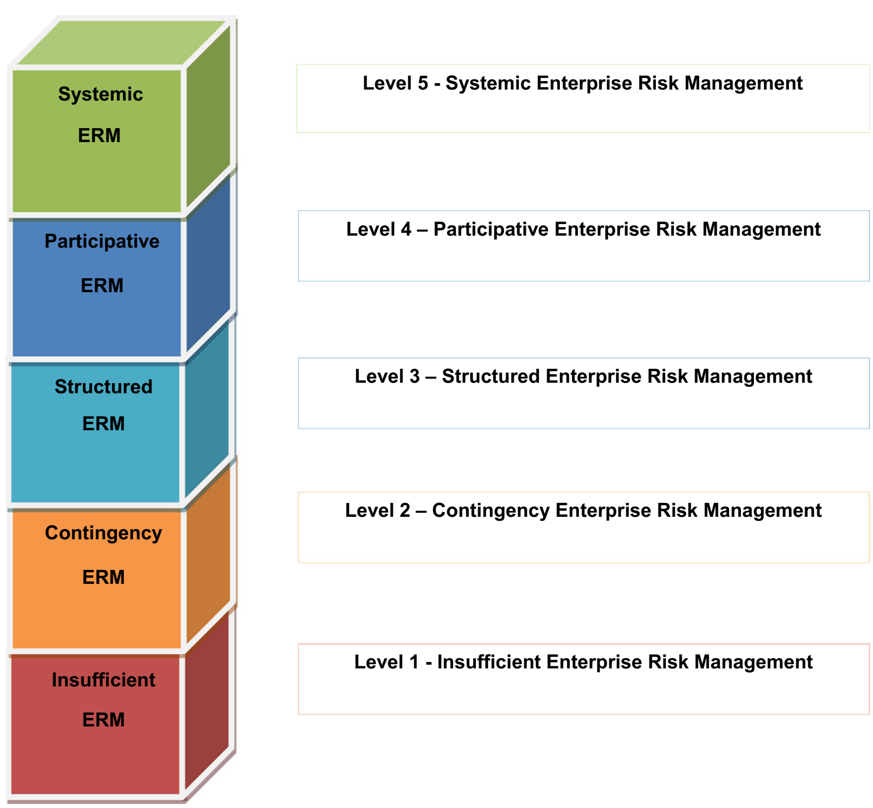


Fig. 4. Level of maturity in enterprise risk management.
Source: Author (2014).

Level 5, called *systemic* enterprise risk management, is the highest level of the classification. In this level, companies have a conscious, organized and transparent enterprise risk management. These companies use external support from consulting firms, partners and research institutes to improve risk management. In addition, the risk management of company increasingly includes the assessment of the risks of its environment of value, considering that the risks do not respect the organizational boundaries, they are sovereign with respect to property boundaries. Thus, it is expected that modern enterprise risk management transcends its practices beyond the boundaries of the organization.

5. Conclusions

The development of this study was dedicated to the analysis of enterprise risk management in the supply chain of large Brazilian companies. The importance of this subject can be assessed by its constant presence on the agenda of managers. Modern society is a source of risks, but also a source to repair them. Humankind dominates the risk, transforming the passive fear by managing it.

Seeking to meet the objectives of the study and answer the research questions, we developed a survey with over 150 companies ranked as the largest Brazilian companies, according to a publication specialized in economics. The statistical analysis was performed through the descriptive analysis of data and then by the multivariate analysis, which allowed us to further deepen the study to propose a solution to the research questions and meet the objectives of the study.

The model proposed for the analysis of enterprise risks presents some points of distinction with respect to the current models. First, we should highlight that the model proposed assesses the enterprise risks by three levels: organization, environment of value and business environment. The organization is exposed to forces and events arising out of these three environments, which have distinctive characteristics, in such a way that the segmented analysis enables higher accuracy. Another important aspect is that the analysis of enterprise risks should transcend the organizational boundaries, that is, in addition to assessing the analysis of the organization's risks in the three levels, it should assess the risks to which the organization's relationships with other agents of the business environment are subject. In other words, the enterprise risk analysis should be a relational analysis, not only of the positioning of the organization, but above all, the enterprise risk analysis must be established in the context of its environment of value, a concept coined in this thesis, which allows the assessment not only of the risks of the main agent, but the organization and the risks arising from the transactions with other agents that generate value. It is worth noting that the enterprise risks do not respect the boundaries of the organization. Concentrating the enterprise risk analysis within the organizational scope might at first strengthen the aspects of efficiency, however, in the medium and long-term, the organization may lose its effectiveness in achieving its strategic objectives.

As a specific goal, which was to identify the enterprise risk management practices, the research has shown the form, frequency, technicality, structure, transparency, participation, communication and the involvement of third parties as elements of the enterprise risk management of large Brazilian companies. Understanding these issues allowed the identification of the main characteristics of the risk management of companies, as well as the segmentation of companies, according to the way they conduct their risk management. In such a way that it is possible to develop a structured way of assessing how large companies deal with enterprise risk management.

The model proposed for the analysis of maturity in enterprise risk management stands out for the originality of the subject itself. Few maturity models focused on enterprise risk management were found in the literature. In addition, the severity and the voracity with which the subject is posed have shown the lack of adherence on certain aspects of current models. Some aspects of distinction of the model proposed should be highlighted. First, it can be seen that the level of risk management assumes higher levels, in such a way that it is no longer a specialty within the organization, but it permeates the key processes of the organization. It was found that companies in the intermediate level of maturity have an enterprise risk management with a high degree of organization, use of methods and techniques, and greater decentralization, that is, distinctive characteristics in the past literature, today they are common to the companies. In addition, the research revealed important new characteristics to assess the level of maturity, including transparency in the communication of potential risks, participation of external agents in risk management, risk assessment in the environment of value.

Acknowledgments

Faculdade de Economia, Administração e Contabilidade-FEA.
Universidade de São Paulo-USP.
Fundação Instituto de Administração-FIA.

References

Aqlan, F., Lam, S.S., 2015a. A fuzzy-based integrated framework for supply chain risk assessment. Int. J. Prod. Econ. 161, 54–63.
Aqlan, F., Lam, S.S., 2015b. Supply Chain risk modeling and mitigation. Int. J. Prod. Res. 53 (18), 5640–5656.
AON, 2012. Aon Risk Management, available at: ⟨http://www.aon.com⟩, accessed: 03/2012.
Bachmann, R., 2001. Trust, power and control in trans-organizational relations. Organ. Stud. 22 (2), 337–365.
Ballou, R.H., 2004. Business Logistics Management: Planning, Organizing, and Controlling the Supply Chain. Pearson Prentice Hall, New York.
Barzel, Y., 2002. Organizational forms and measurement costs. J. Inst. Theor. Econ. 161 (3), 357–373.
Bateman, T.S., Snell, S.A., 2012. Management: Leading & Collaborating in the Competitive World Management. McGraw-Hill Higher Education, New York.
Beasley, M.S., Clune, R., Hermanson, D.R., 2005. Enterprise risk management: an empirical analysis of factors associated with the extent of implementation. J. Account. Public Policy 24, 521–531.
Blome, C., Schoenherr, T., 2011. Supply risk management in financial crises: a multiple case study approach. Int. J. Prod. Econ. 134 (1), 43–57.
Chen, J., Sohal, A.S., Prajogo, D.I., 2013. Supply chain operational risk mitigation: a collaborative approach. Int. J. Prod. Res. 51 (7), 2186–2199.
Chopra, S., Sodhi, M.S., 2004. Managing Risk to Avoid Supply-Chain Breakdown. MIT Sloan Management Review. 46 (1), 53–61.
Christopher, M., Peck, H., 2004. Building the resilient supply chain. Int. J. Logist. Manag. 15 (2), 1–14.
Christopher, M., Holweg, M., 2011. Supply chain 2.0: managing supply chains in the era of turbulence. Int. J. Phys. Distrib. Logist. Manag. 41 (1), 63–82.
Coase, R.H., 1937. The nature of the firm. Econ. New Ser. 4 (16), 386–405.
Coase, R.H., 1960. The problem of social cost. J. Law Econ. 3, 01–44.
Coase, 2014. available at: ⟨http://www.coase.org⟩, accessed: 03/2014.
Cooper, D.C., Schindler, P.S., 2001. Business Research Methods. McGraw-Hill, New York.
COSO, 2004. Committee of Sponsoring Organizations of the Treadway Commission. Enterprise Risk Management-Integrated Framework.
Damodaran, A., 2008. Strategic Risk Taking: A Framework for Risk Management. Pearson Prentice Hall, New York.
Deloitte, 2005. Destruidores de valor. Rev. Mundo Corp. 3 (10), 7–9.
Deloitte, 2013. The Ripple Effect: How Manufacturing and Retail Executives View the Growing Challenge of Supply Chain Risk, available at: ⟨http://deloitte.com⟩, accessed: 07/2014.
Denzin, N., 1970. Strategies of multiple triangulation. In: Denzin, N. (Ed.), The Research Act in Sociology: A Theoretical Introduction to Sociological Method. Butterworth, London.
Ernest and Young, 2012. available at: ⟨http://ey.com⟩, accessed: 05/2012.

Espejo, R., 1993. Management of Complexity in Problem Solving, in: Raul Espejo e Markus Schwaninger, Organisational Fitness: Corporate Effectiveness Through Management Cybernetics. Campus Verlag, New York.
Espejo, R., et al., 1996. Giving Requisite Variety To Management: A Discussion Based on the Viable System Model, In: Organizational Transformation And Learning. Wiley & Sons, Ontario.
EXAME, 2011. Melhores & Maiores: As 1.000 Maiores Empresas do Brasil. Revista Exame, Edição Especial, São Pailo.
FERMA, 2003. Federation of European Risk Management Association. Normas de gestão de riscos, available at: ⟨http://www.ferma.eu⟩, accessed: 03/2012.
Gaultier-Gaillard, S., Louisot, J.P., Rayner, J., 2009. Managing reputational risk – From theory to practice, in: Reputation Capital: Building and Maintaining Trust in the 21st Century. Springer Berlin Heidelberg, Berlin.
Grey, W., Shi, D., 2005. Enterprise risk management: a value chain perspective. handbook of integrated risk management for e-business. J. Ross Publishing, Florida.
Habermann, M., Blackhurst, J., Metcalf, A.Y., 2015. Keep your friends close? supply chain design and disruption risk. Decis. Sci. 46 (3), 491–526.
Hahn, G.J., Kuhn, H., 2012. Value-based performance and risk management in supply chains: a robust optimization approach. Int. J. Prod. Econ. 139, 135–144.
Hair Jr., J.F., Anderson, R.E., Tatham, R.L., Black, W.C., 1998. Multivariate Data Analysis.. Prentice Hall, Upper Saddle River, New Jersey.
Hauck Jr., W.W., Donner, A., 1977. Wald's test as applied to hypotheses in logit analysis. J. Am. Stat. Assoc. 72 (360), 851–853.
Heckmann, I., Comes, T., Nickel, S., 2015. A critical review on supply chain risk-Definition, measure and modeling. Omega 52 (C), 119–132.
Hillson, D.A., 1997. Towards a Risk Maturity Model. Int. J. Proj. Bus. Risk Manag. 1 (1), 35–45.
Hitt, M., Ireland, R.D., Hoskisson, R., 2014. Strategic Management: Concepts and Cases: Competitiveness and Globalization. Cengage Learning, Boston.
Hofmann, H., Busse, C., Bode, C., Henke, M., 2014. Sustainability-Related Supply Chain Risks: Conceptualization and Management. Bus. Strategy Environ. 23, 160–174.
Hosmer, D.W., Lemshow, S., 2000. Applied Logistic Regression. Wiley, New York.
Hoyt, R.E., Liebenberg, A.F., 2011. The value of enterprise risk management. J. Risk Insur. 78 (4), 795–822.
Hult, G.T.M., Craighead, C.W., Ketchen, D.J., 2010. Risk uncertainty and supply chain decisions: a real options perspective. Decis. Sci. 41 (3), 435–458.
IBGC, 2007. Instituto Brasileiro de Governança Corporativa. Guia de orientação para o gerenciamento de riscos corporativos, available at: ⟨http://www.ibgc.org.br⟩, accessed: 06/2012.
IFAC, 2001. International Federation of Accountants. Governance in the Public Sector: A Governing Body Perspective, available at: ⟨http://www.ifac.org⟩, accessed: 09/2012.
IMA, 1997. Institute of Management Accountants. Measuring and Managing Shareholder Value Creation, available at: ⟨http://www.imanet.org⟩, accessed: 10/2012.
ISO 31000, 2009. International Organization for Standardization. ISO 31000: Risk Management – Principles and Guidelines on Implementation.
Juttner, U., 2005. Supply chain risk management. Int. J. Logist. Manag. 16, 120–141.
Juttner, U., Peck, H., Christopher, M., 2003. Supply chain risk management: outlining and agenda for future research. Int. J. Logist.: Res. Appl. 6 (4), 17–32.
Kim, C., Mauborgne, R., 2004. Blue Ocean Strategy. Harv. Bus. Rev., 76–85.
Lee, H.L., 2004. The triple-a supply chain. Harv. Bus. Rev. 83, 102–112.
Li, G., Fan, H., Lee, P.K.C., Cheng, T.C.E., 2015. Joint supply chain risk management: an agency and collaboration perspective. Int. J. Prod. Econ. 164, 83–94.
Louisot, J.P., 2010. Gestion des risques. Afnor, Paris.
Manuj, I., Mentzer, J.T., 2008. Global supply chain risk management. J. Bus. Logist. 29 (1), 133–156.
Manuj, I., Esper, T.L., Stank, T.P., 2014. Supply chain risk management approaches under different conditions of risk. J. Bus. Logist. 35 (3), 241–258.
McShane, M.K., Nair, A., Rustambekov, E., 2010. Does Enterprise Risk Management increase Firm Value? J. Account. Audit. Financ. 26 (4), 641–658.
Ménard, C., 1995. Markets as institutions versus organizations as markets? Disentangling some fundamental concepts. J. Econ. Behav. Organ. 28 (2), 161–182.
Ménard, C., 1997. Le pilotage des formes organisationnelles hybrids. Rev. Econ. 48 (3), 741–750.
Ménard, C., 2000. Enforcement Procedures and Governance Structures: What Relationship?. In: Ménard (Ed.), Institutions Contracts and Organizations. Edward Elgar, Cheltenham.
Mikes, A., 2009. Risk management and calculative cultures. Manag. Account. Res. 20, 18–40.
Miller, R., Lessard, D., 2001. Understanding and managing risks in large engineering projects. Int. J. Proj. Manag. 19, 437–443.
Mintzberg, H., Ahlstrand, B., Lampel, J., 2005. Strategy Safari: A Guided Tour Through The Wilds of Strategic Management. Simon and Schuster, New York.
Nee, V., 2003. New Institutionalism, Economic and Sociological. Handbook for Economic Sociology. Princeton University Press, Princenton.
Neiger, D., Rotaru, K., Churilov, L., 2009. Supply chain risk identification with value-focused process engineering. J. Oper. Manag. 27 (2), 154–168.
Nooraie, S.V., Parast, M.M., 2015. A multi-objective approach to supply chain risk management: Integrating visibility with supply and demand risk. Int. J. Prod. Econ. 161, 192–200.
Norrman, A., Jansson, U., 2004. Ericsson's proactive supply chain risk management approach after a serious sub-supplier accident. Int. J. Phys. Distrib. Logist. Manag. 34 (5), 434–456.
North, D.C., 1990. Institutions, Institutional Change and Economic Performance. Cambridge University Press, Cambridge.
Olson, D.L., Swenseth, S.R., 2014. Trade-offs in supply chain system risk mitigation. Syst. Res. Behav. Sci. 31, 565–579.
Paape, L., Speklé, R., 2012. The adoption and design of enterprise risk management practices: an empirical study. Eur. Account. Rev. 21 (3), 533–564.
Pestana, M.H., Gageiro, J.N., 2000. Análise de Dados para Ciências Sociais: a complementaridade do SPSS. Edições Sílabo, São Paulo.
Porter, M., 2008. Competitive Advantage: Creating and Sustaining Superior Performance. Free Press, New York.
Porter, M., 2004. Competitive Strategy: Techniques for Analyzing Industries and Competitors. Free Press, New York.
Porter, M., Millar, V.E., 1985. How information gives you competitive advantage. Harv. Bus. Rev., 149–174.
RIMS, 2006. Risk and Insurance Management Society. RIMS Risk Maturity Model (RMM) for Enterprise Risk Management, available at: ⟨http://www.rims.org⟩, accessed: 04/2012.
Shank, J.K., Govindarajan, V., 1993. Strategic Cost Management: The New Tool For Competitive Advantage. The Free Press, New York.
Sodhi, M.S., Son, B., Tang, C.S., 2012. Researchers' perspectives on supply chain risk management. Prod. Oper. Manag. 21 (1), 1–13.
Sydow, J., Frenkel, S.J., 2013. Labor, risk, and uncertainty in global supply networks-exploratory insights. J. Bus. Logist. 34 (3), 236–247.
Tang, C.S., 2006a. Perspectives in supply chain risk management. Int. J. Prod. Econ. 103 (2), 451–488.
Tang, C.S., 2006b. Robust strategies for mitigating supply chain disruptions. Int. J. Logist.: Res. Appl. 9 (1), 33–45.
Tang, C.S., Tomlin, B., 2008. The power of flexibility for mitigating supply chain risks. Int. J. Prod. Econ. 116 (1), 12–27.
Thun, J.H., Hoenig, D., 2011. An empirical analysis of supply chain risk management in the German automotive industry. Int. J. Prod. Econ. 131 (1), 242–249.
Veiga, J.E. da, 2005. Do Global ao Local. Armazém do Ipê, Campinas.
Walters, D., Rainbird, M., 2004. The demand chain as an integral component of the value chain. J. Consum. Mark. 21 (7), 465–475.
Walters, D., 2007. Supply chain Risk Management: Vulnerability and Resilience in Logistics. Ed. Kogan-page, London.
Williamson, O.E., 1975. Markets and Hierarchies: Analysis and Antitrust Implications. The Free Press, New York.
Williamson, O.E., 1985. The Economic Institutions of Capitalism: Firms, Markets, Relational Contracting. The Free Press, New York.
Zsidisin, G.A., 2003. Managerial perceptions of supply risk. J. Supply Chain Manag. 39 (1), 14–25.